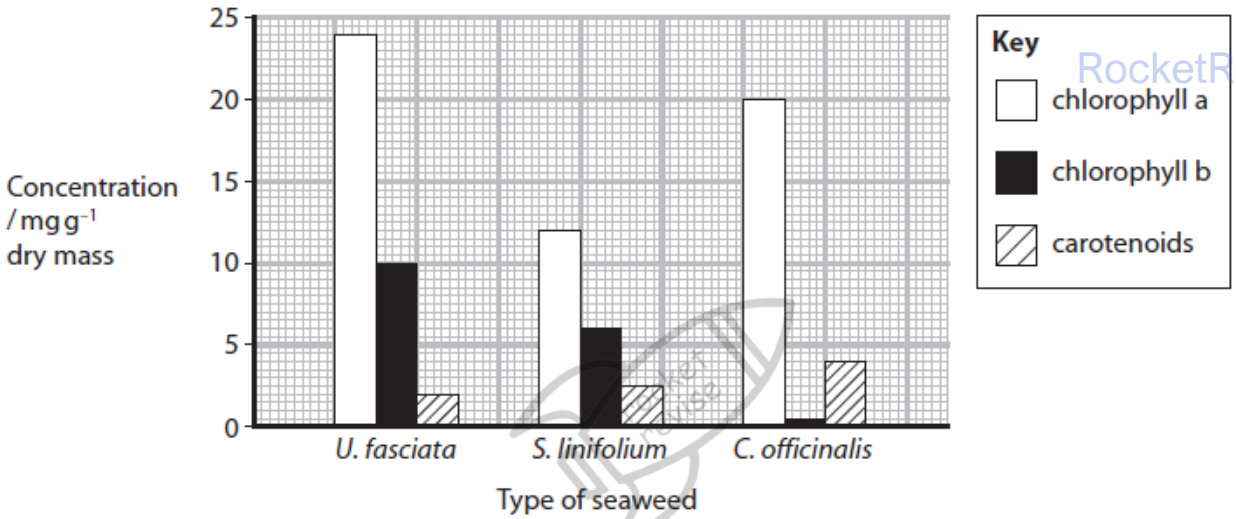
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Questions

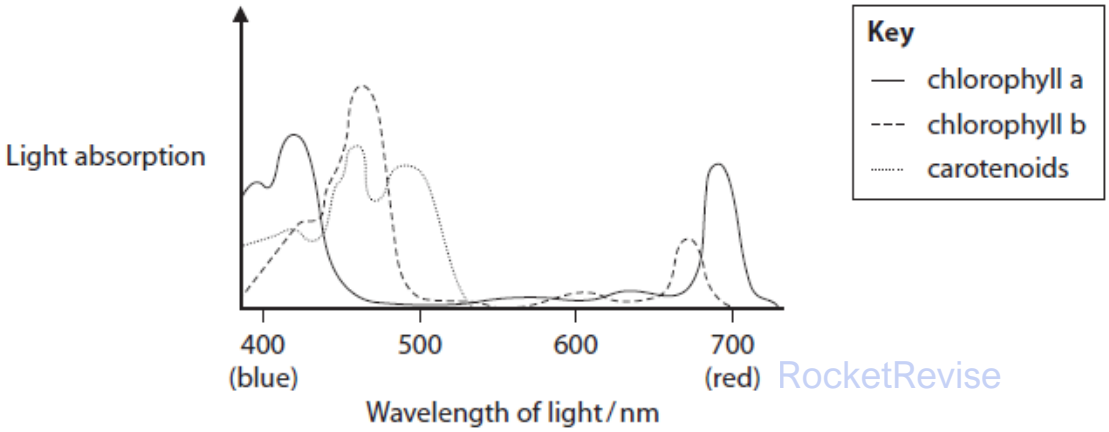
Q1.

The composition of three seaweeds, *U. fasciata*, *S. linifolium* and *C. officinalis*, collected from a beach in Egypt, was determined.

(a) The graph shows the chloroplast pigments extracted from these seaweeds.



The graph shows the absorption spectrum for these three chloroplast pigments.



(i) Which row of the table shows the colour of each of these seaweeds?

(1)

	U. fasciata	*S. linifolium*	*C. officinalis*
☒ **A**	green	brown	red
☒ **B**	green	red	green
☒ **C**	red	brown	red
☒ **D**	red	green	green

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(ii) Explain the role of chloroplast pigments in the light-dependent reactions of photosynthesis.

(2)

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(b) The table shows some of the compounds found in two of these seaweeds.

Compound	*U. fasciata*	*S. linifolium*
*carotenoids / mg g^{-1} dry mass	2.00	3.00
*phenols / a.u. g^{-1} dry mass	11.95	10.35
*flavonoids / a.u. g^{-1} dry mass	7.04	4.53
carbohydrates / percentage of dry mass	23.70	25.03
proteins / percentage of dry mass	9.56	14.89
lipids / percentage of dry mass	2.96	2.16

* antioxidants

(i) Describe how GALP, produced in chloroplasts, is converted into protein.

(3)

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*(ii) These seaweeds are considered to be a healthy, environmentally-friendly and economic food source.

Comment on this statement.

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Use the information in the table and your own knowledge to support your answer.

(6)

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(Total for question = 12 marks)

Q2.

The following equation summarises photosynthesis.

$$6CO_2 + 6H_2O \longrightarrow C_6H_{12}O_6 + 6O_2$$

The diagrams show the bonds in carbon dioxide, water and oxygen.

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carbon dioxide	water	oxygen
O=C=O	H–O–H	O=O

Energy is needed to break chemical bonds, and to form new chemical bonds. This is called the bond energy.

The table shows some bond energies for the bonds in carbon dioxide, water, glucose and oxygen.

Type of bond	Bond energy / kJ per bond
C=O	785
O–H	462
O=O	487

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(a) (i) In photolysis, one of the bonds in each water molecule is broken.

Using the equation for photosynthesis, calculate how much energy is released by photolysis in order for one molecule of glucose to be made.

(1)

Answer ..... kJ

(ii) Explain how light energy is converted into chemical energy in the form of ATP.

(4)

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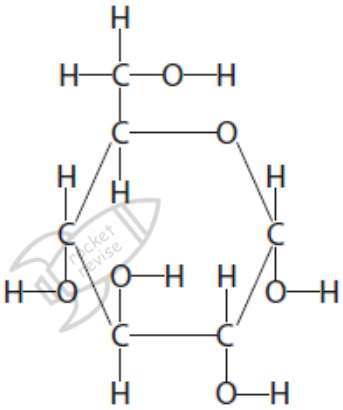
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(b) The diagram shows the structure of glucose.

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The energy needed to form the bonds in one molecule of glucose is 9164 kJ.

(i) State what other information is needed in the table of bond energies for this value to be calculated.

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(1)

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(ii) Where in the chloroplasts are these bonds formed?

(1)

☒ **B** matrix

☒ **B** matrix

☒ **C** stroma

☒ **D** thylakoid membrane

(c) Glucose is used by plants in the production of amino acids.

(i) Which row of the table describes how two amino acids join together?

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(1)

	Bond formed between	Type of reaction
☒ **A**	carbon and nitrogen	condensation
☒ **B**	carbon and nitrogen	hydrolysis
☒ **C**	oxygen and nitrogen	condensation
☒ **D**	oxygen and nitrogen	hydrolysis

(ii) Explain why amino acids cannot be produced from glucose alone.

(2)

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(Total for question = 10 marks)

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Q3.

Chloroplasts are involved in both the light-dependent reactions and light-independent reactions of photosynthesis.

(a) Which row of the table shows where the light-dependent reactions and light-independent reactions take place?

(1)

	Light-dependent reactions	Light-independent reactions
☒ **A**	stroma	stroma
☒ **B**	stroma	thylakoid membranes
☒ **C**	thylakoid membranes	stroma
☒ **D**	thylakoid membranes	thylakoid membranes

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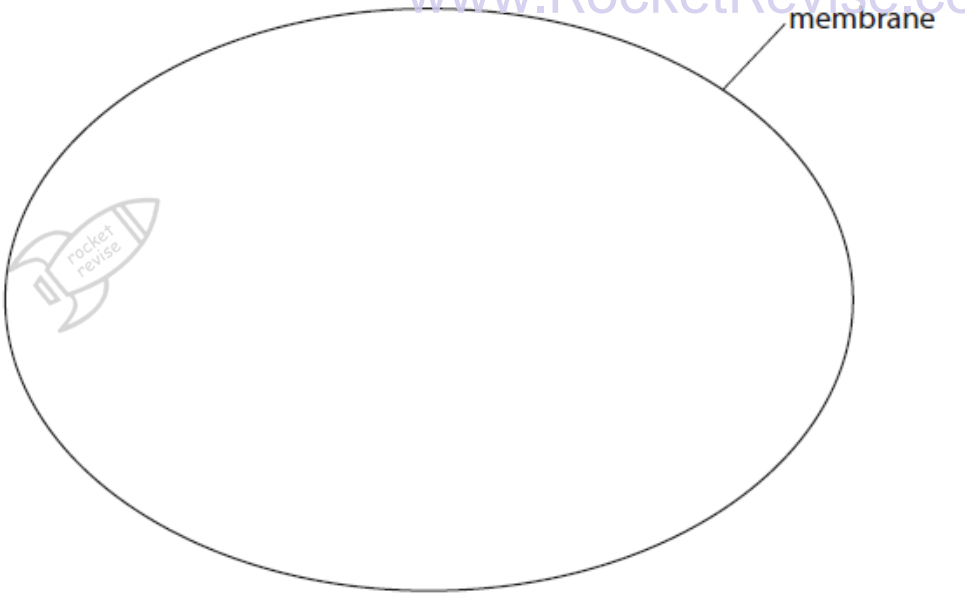
(b) The diagram shows the outline of a chloroplast.

Draw **three** labelled features on this diagram that are found in a chloroplast, other than the stroma and the thylakoid membranes.

(3)

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(c) An absorption spectrum shows how much light is absorbed by chloroplasts at different wavelengths of light.

The table shows the colour of light at four wavelengths.

Wavelength of light / nm	460	520	600	680
Colour of light	blue	green	yellow	red

Which wavelength of light is absorbed the **least** by chloroplasts?

(1)

☒ **A** 460 nm

☒ **B** 520 nm

☒ **C** 600 nm

☒ **D** 680 nm

(d) State what is meant by the term **action spectrum**.

RocketRevise

(1)

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(e) Chloroplast pigments can be separated and then identified by their Rf values.

(i) Which process can be used to separate chloroplast pigments?

(1)

☒ **B** dendrochronology

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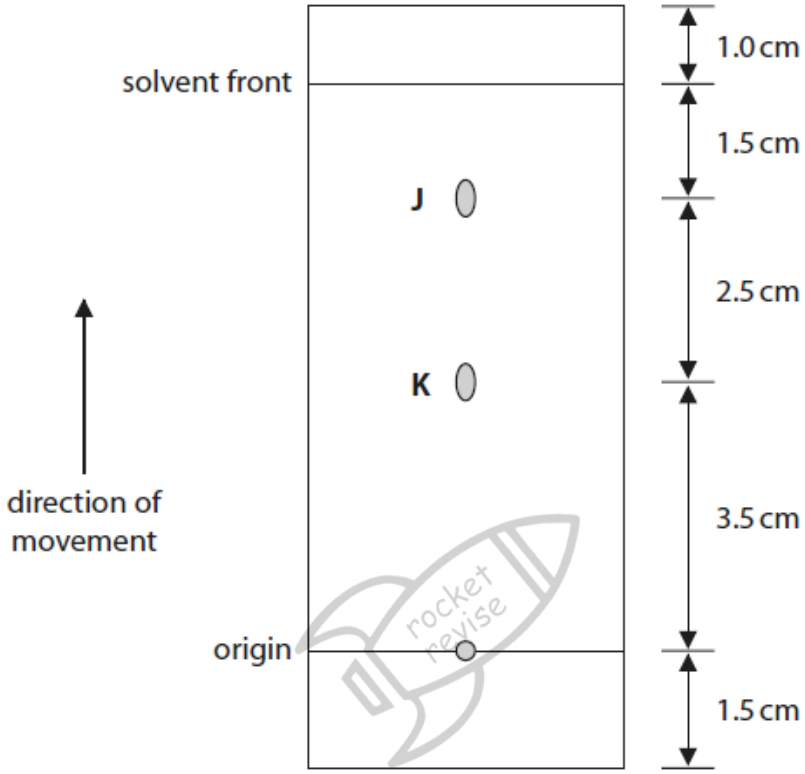
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☒ **B** dendrochronology

☒ **C** osmosis

☒ **D** PCR

(ii) The diagram shows separated chloroplast pigments, **J** and **K**.



What is the Rf value for chloroplast pigment **J**?

(1)

☒ **A** 0.625

☒ **B** 0.800

☒ **C** 0.830

☒ **D** 1.714

(Total for question = 8 marks)

Q4.

Chloroplasts and mitochondria are two organelles found in some plant cells.

(a) Compare and contrast the structure of a chloroplast with the structure of a mitochondrion.

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(3)

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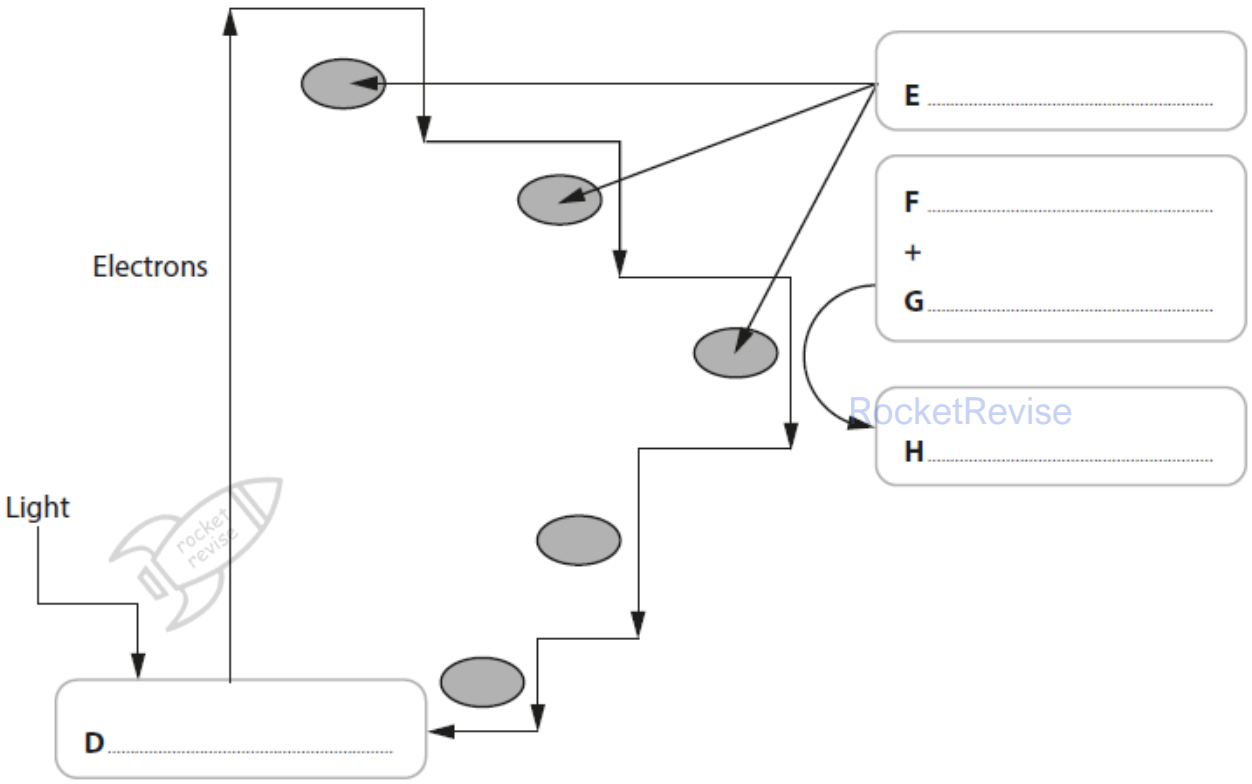
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(b) The light-dependent reactions of photosynthesis can result in either cyclic or non-cyclic photophosphorylation.

(i) The diagram shows cyclic photophosphorylation.

Complete the diagram by writing the correct word or words on the dotted lines, labelled D, E, F, G and H.

(3)



(ii) Name **one** molecule that is produced in non-cyclic photophosphorylation that is **not** produced in cyclic photophosphorylation.

(1)

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(iii) Describe the role of photolysis in non-cyclic photophosphorylation.

(2)

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(Total for question = 9 marks)

Q5.

Seaweeds are a group of organisms that carry out photosynthesis.

Identifying the types and proportions of chlorophyll pigments is important in the classification of seaweeds and biodiversity studies.

(a) State the meaning of the term **biodiversity**.

(2)

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(b) Explain the role of chlorophyll in the light-dependent reactions of photosynthesis.

(2)

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(c) An investigation compared the effectiveness of two solvents, propanone and DMSO, in extracting photosynthetic pigments.

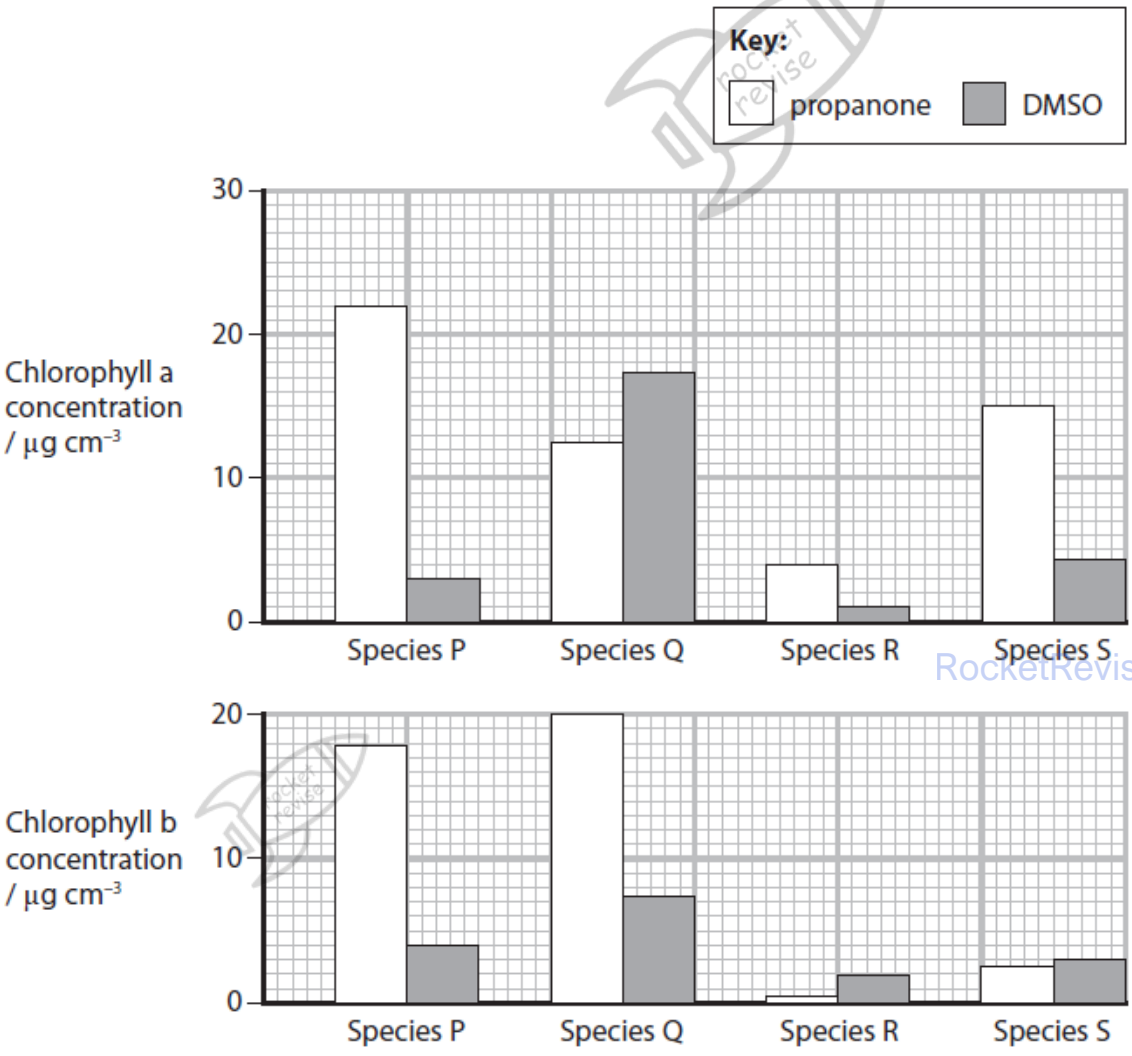
Four species of seaweed were collected from the Indian coast of Tamil Nadu and taken to the laboratory where the species were identified.

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Each species of seaweed was split into two samples of equal mass.

The photosynthetic pigments were extracted from each sample using one of the two solvents and their concentrations determined.

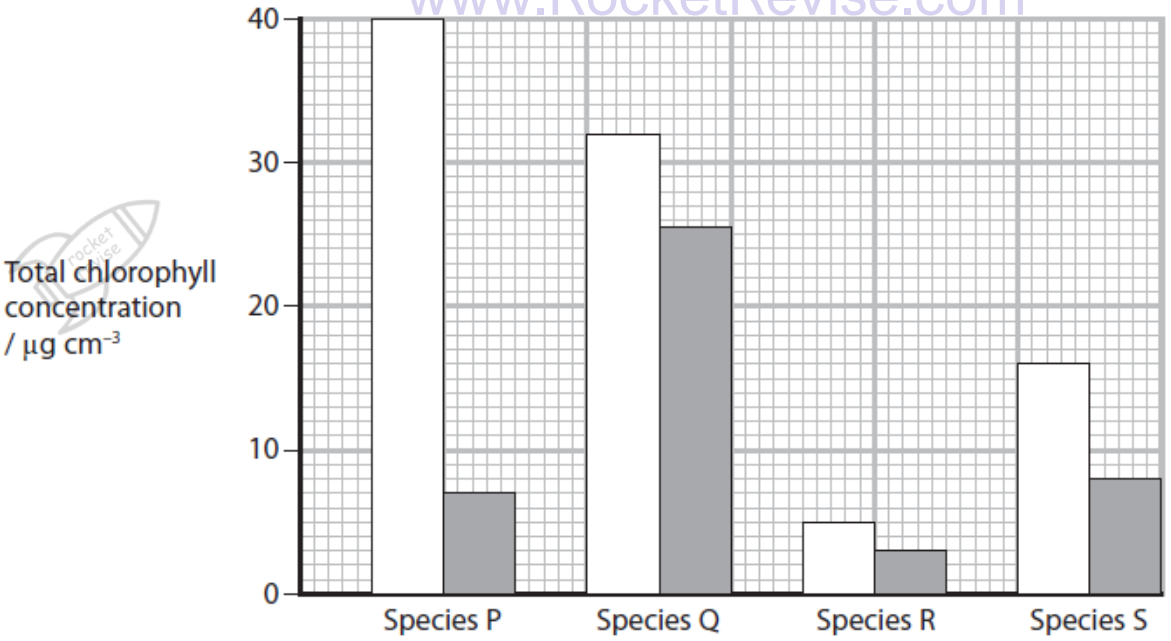
The graphs show some of the results of this investigation.



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* (i) Discuss the effectiveness of these extraction methods in the identification of these species.

Use the information in the graphs to support your answer.

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(ii) Suggest why different concentrations of chlorophyll were obtained using these two different solvents.

(2)

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(Total for question = 12 marks)

Q6.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Photosynthesis consists of the light-dependent reactions and the light-independent reactions.

(a) In the light-dependent reactions, light energy is converted into energy stored in ATP.

(i) Explain why light energy is converted into energy stored in ATP.

(2)

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(ii) An equation can summarise the production and breakdown of ATP.

Complete this equation, by writing the names of the substrates and the type of reaction on the three dotted lines provided.

(2)



(iii) Explain the role of light energy in the light-dependent reactions.

(3)

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(b) The light-independent reactions use products of the light-dependent reactions to produce simple sugars.

(i) Which row of the table shows the products of the light-dependent reactions that are used in the light-independent reactions?

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(1)

	ATP produced by	NADP
☐ **A**	cyclic photophosphorylation	oxidised
☐ **B**	cyclic photophosphorylation	reduced
☐ **C**	non-cyclic photophosphorylation	oxidised
☐ **D**	non-cyclic photophosphorylation	reduced

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(ii) Simple sugars have the formula $C_nH_{2n}O_n$.

Name the inorganic molecule from which each element in a simple sugar originated.

(2)

C .....

H .....

O .....

(iii) Simple sugars are used in the synthesis of new biological molecules.

Which row of the table shows the inorganic ions that are needed to synthesise these new biological molecules?

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(3)

New biological molecule	Nitrates	Phosphates	Both nitrates and phosphates	Neither nitrates nor phosphates
protein	☐	☐	☐	☐
RNA	☐	☐	☐	☐
triglyceride	☐	☐	☐	☐

(Total for question = 13 marks)

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Q7.

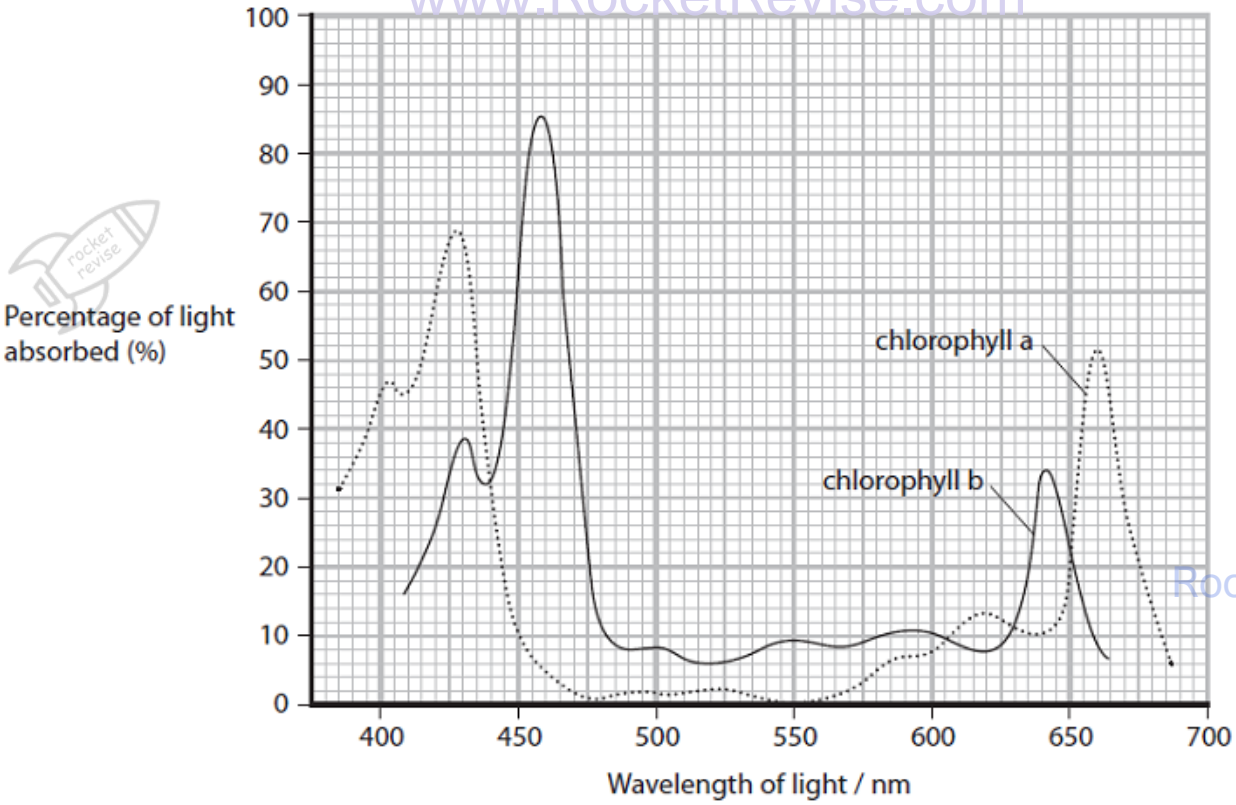
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Thylakoid membranes contain chloroplast pigments and molecules of ATP synthase. These membranes are the site of the light-dependent reactions.

(a) The graph shows the absorption spectrums of two chloroplast pigments from one plant.

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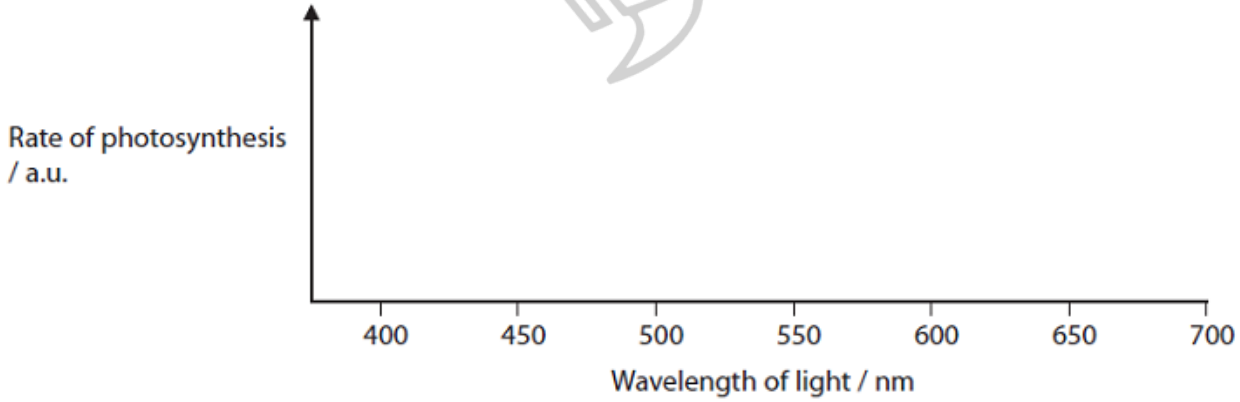
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(i) Draw the action spectrum for this plant on the axes below.

(2)



(ii) Calculate how many times greater the percentage of light absorbed by chlorophyll b is than that absorbed by chlorophyll a, at their maximum absorptions.

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(1)

Answer .....

(iii) Thylakoid membranes contain different types of chlorophyll pigment.

Explain the advantage of this for a plant.

(2)

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(b) The ATP synthase molecules are proteins.

(i) Which row of the table describes the bonds that form between amino acids in a protein?

(1)

		Name of bond	Parts of the amino acid that the bond forms between
☒	**A**	peptide	amino group and carboxyl group
☒	**B**	peptide	carboxyl group and R group
☒	**C**	phosphodiester	amino group and carboxyl group
☒	**D**	phosphodiester	carboxyl group and R group

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(ii) Products of the light-independent reactions are used to synthesise amino acids.

Describe how plants synthesise amino acids.

(2)

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(Total for question = 8 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Chloroplasts are involved in both the light-dependent reactions and the light independent

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reactions of photosynthesis.

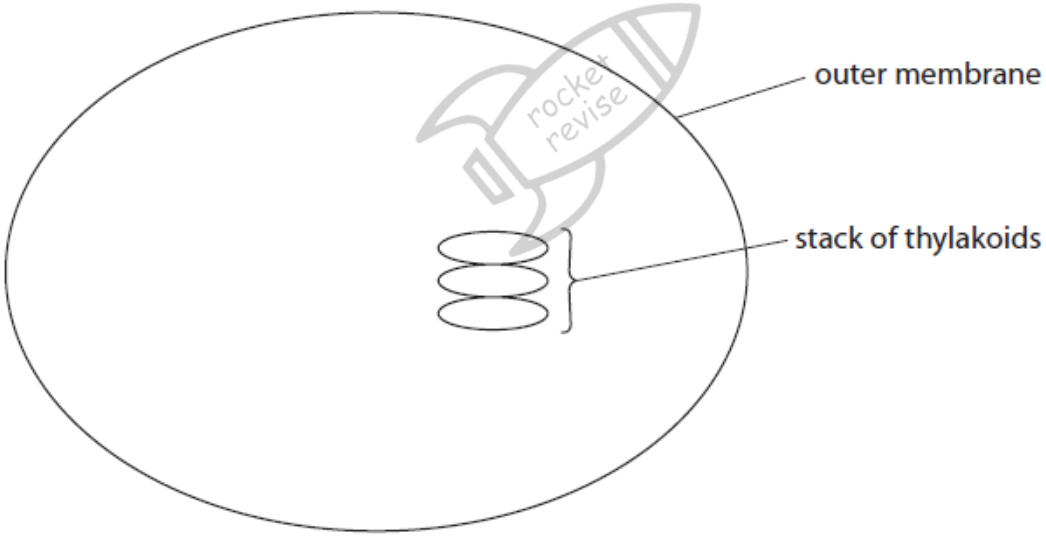
(a) Which row of the table is correct for these two reactions?

(1)

	Light-dependent reactions	Light-independent reactions
☒ A	ADP is phosphorylated, releasing energy	ATP is hydrolysed, requiring energy
☒ B	ADP is phosphorylated, requiring energy	ATP is hydrolysed, releasing energy
☒ C	ATP is hydrolysed, releasing energy	ADP is phosphorylated, requiring energy
☒ D	ATP is hydrolysed, requiring energy	ADP is phosphorylated, releasing energy

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(b) The diagram shows part of a chloroplast.



(i) Complete this diagram to show **three** other labelled structures found in a chloroplast.

(3)

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(ii) Compare and contrast the structure of the outer membrane of a chloroplast with that of a thylakoid membrane.

(4)

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(Total for question = 8 marks)

Q9.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) The table gives some information about structures found in chloroplasts.

Structure	Function	Other information
P	protein synthesis	some are 20 nm in size
Q	site of light-independent reactions	colourless
R	compartment for accumulation of hydrogen ions	largest are 4.35×10^{-4} mm long
S	storage granule	1 to 35 μm in size

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(i) Where in the chloroplast is RUBISCO active?

(1)

☒ **A** P

☒ **B** Q

☒ **C** R

☒ **D** S

(ii) What is stored in S?

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(1)

- ☒ **A** GALP
- ☒ **B** lipid
- ☒ **C** starch
- ☒ **D** sucrose

(iii) Where is GP formed?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

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(iv) Which is the correct order of size, smallest to largest?

(1)

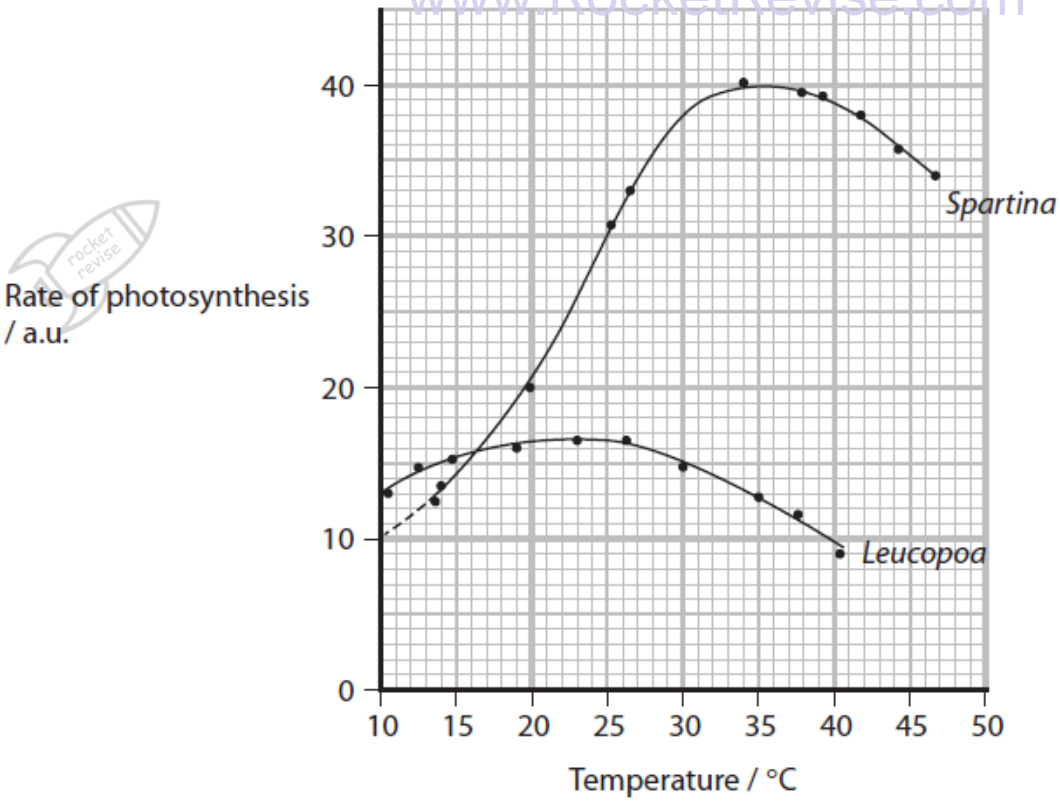
- ☒ **A** P R S
- ☒ **B** P S R
- ☒ **C** R S P
- ☒ **D** S R P

(b) The graph shows the rate of photosynthesis in two types of plant, *Spartina* and *Leucopoa*, at different temperatures.

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(i) Which could be the units for rate of photosynthesis?

(1)

☒ **A** mg CO_2 produced mm^{-2} hr^{-1}

☒ **B** mg CO_2 produced mm^{-1} hr^{-2}

☒ **C** mg CO_2 used mm^{-2} hr^{-1}

☒ **D** mg CO_2 used mm^{-1} hr^{-2}

(ii) Compare and contrast the rate of photosynthesis of these two species of plant, at different temperatures.

(3)

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(iii) Describe how a Q_{10} value for the rate of photosynthesis can be calculated, using this graph.

(2)

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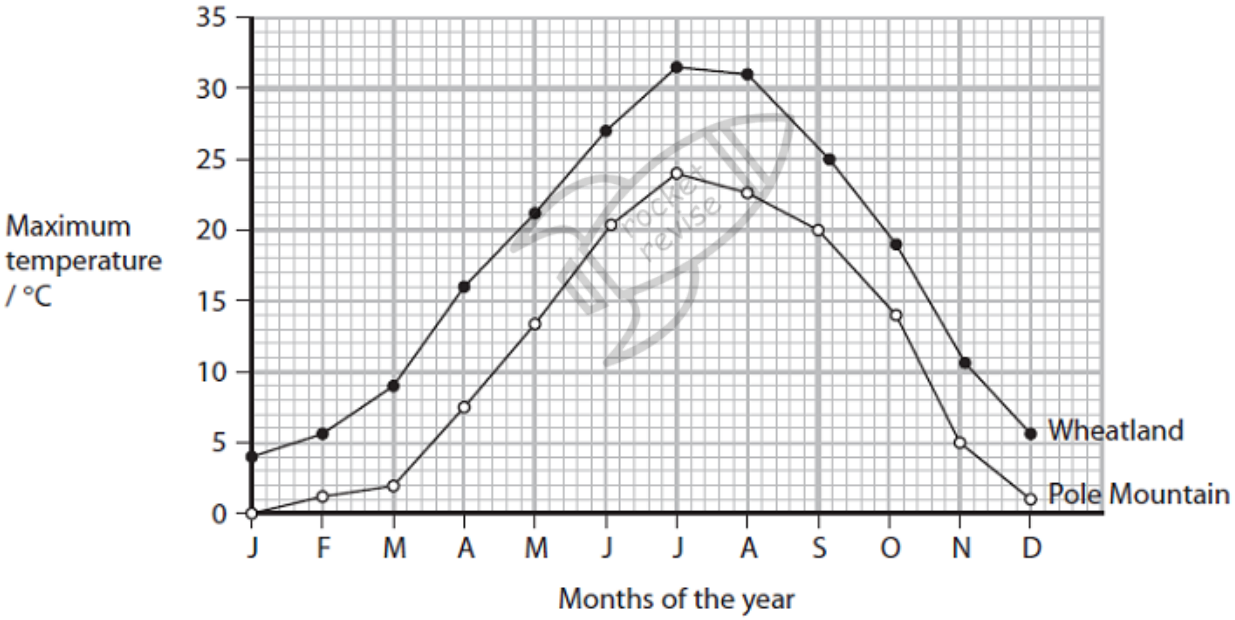
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(iv) The graph shows the maximum temperature, for each month, in two regions of North America: Wheatland and Pole Mountain.



Explain in which of these two regions of North America *Spartina* is most likely to grow.

Use the information in both graphs to support your answer.

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(3)

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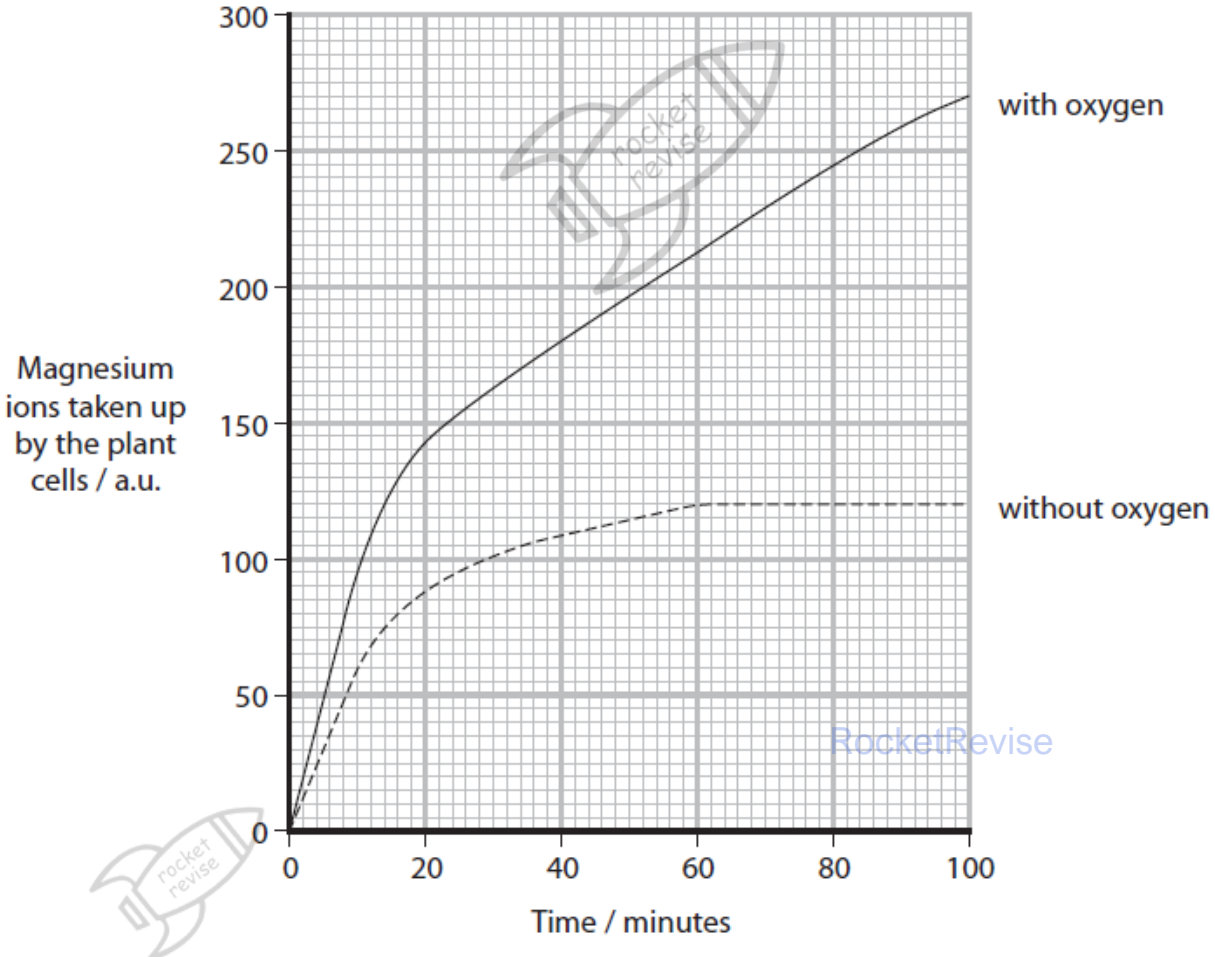
(Total for question = 13 marks)

Q10.

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A student investigated the uptake of magnesium ions by some plant cells in the presence and absence of oxygen.

The graph shows the results of this investigation.



(a) Explain the importance of magnesium ions to a plant.

(2)

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(b) The student suggested that magnesium ions are taken up by active transport.

Explain why the student came to this conclusion.

Use the information in the graph to support your answer.

(4)

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(Total for question = 6 marks)

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Q11.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

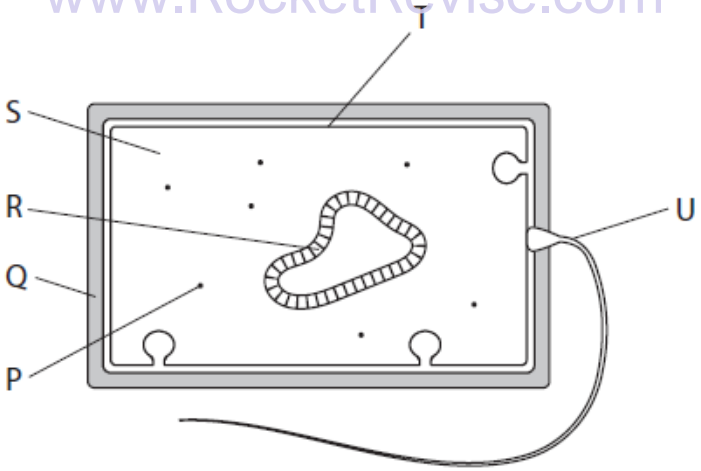
Some bacteria are photosynthetic.

Bacteria do not have chloroplasts.

(a) The diagram shows a photosynthetic bacterium.

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(i) Which labelled structure contains the genes coding for the enzymes involved in photosynthesis?

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(1)

☒ **A** P

☒ **B** Q

☒ **C** R

☒ **D** T

(ii) Where do the light-dependent reactions take place?

(1)

☒ **A** Q

☒ **B** S

☒ **C** T

☒ **D** U

(iii) Where do the light-independent reactions take place?

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(1)

☒ **A** Q

☒ **B** S

☒ **C** T

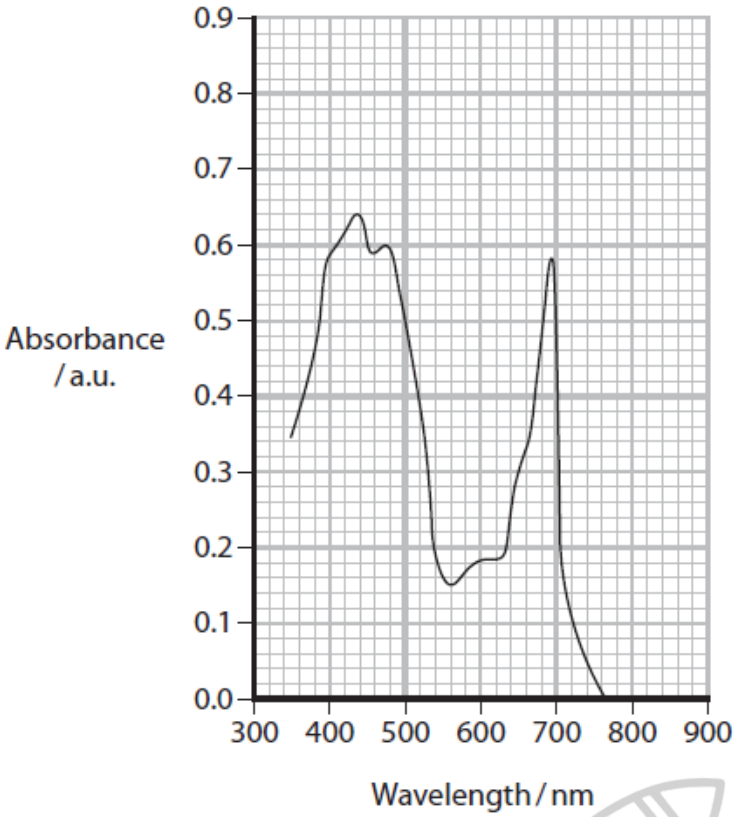
☒ **D** U

(b) The graphs show the absorption spectrum of the photosynthetic pigments in a plant and in a bacterium.

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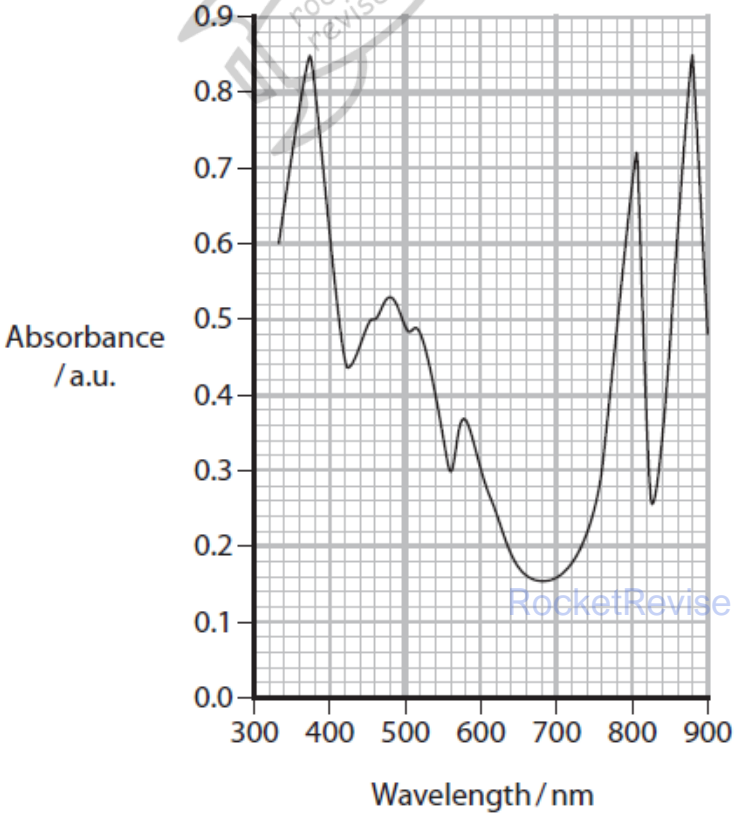
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Plant



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Bacterium



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Describe **two** conclusions that can be made about the differences in absorption spectra shown in these two graphs.

(2)

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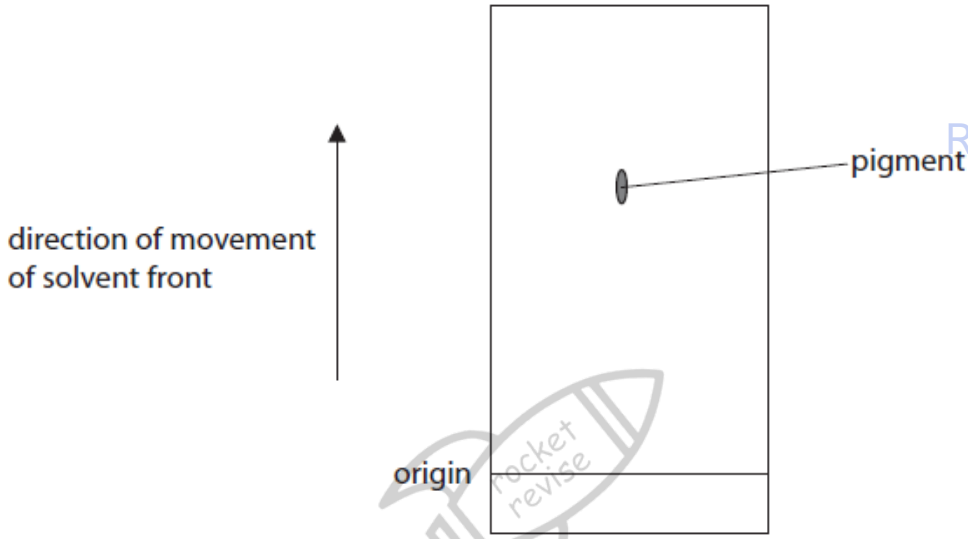
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(c) Photosynthetic pigments can be identified using chromatography.

The diagram shows part of a chromatogram.



Measurements are taken from a chromatogram to determine the Rf value for a photosynthetic pigment.

Describe the measurements taken and how they can be used to calculate the Rf value for this pigment.

(3)

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(Total for question = 8 marks)

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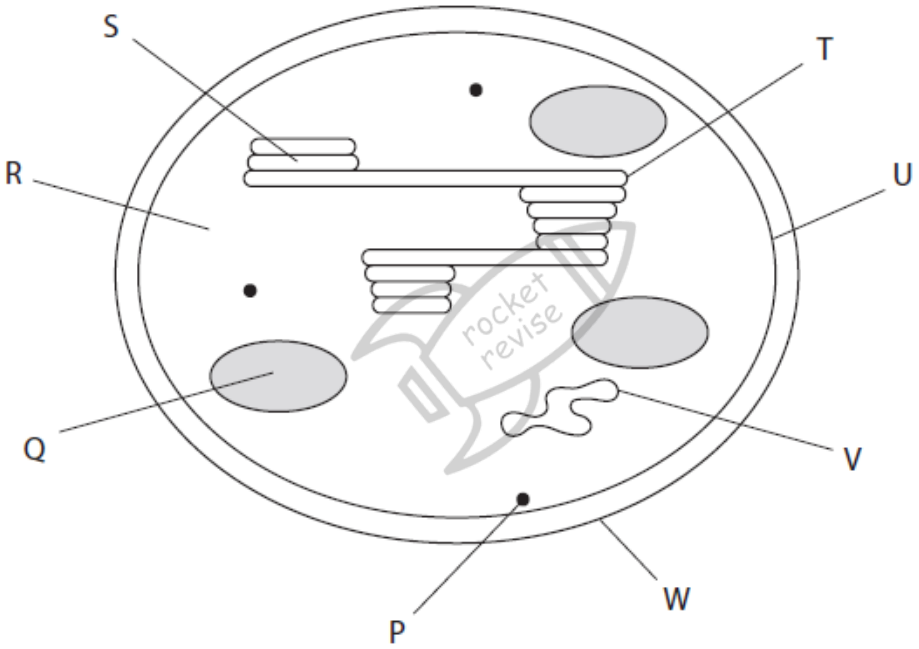
Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of a chloroplast is related to its role in photosynthesis.

The diagram shows a chloroplast.

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(a) Which is the site of photolysis?

(1)

☒ **A** P

☒ **B** R

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☒ **C** T

☒ **D** V

(b) Which contains a gene coding for RUBISCO?

(1)

☒ **A** P

☒ **B** Q

☒ **C** S

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☒ **D** V

(c) Where do hydrogen ions accumulate for use in photosynthesis?

(1)

☒ **A** Q

☒ **B** R

☒ **C** S

☒ **D** V

(d) (i) How many of the structures labelled P, T, U and W are made of phospholipids?

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(1)

☒ **A** 1

☒ **B** 2

☒ **C** 3

☒ **D** 4

(ii) Which describes a phospholipid?

(1)

☒ **A** glycerol attached to two fatty acids and one phosphate group

☒ **B** glycerol attached to two fatty acids and two phosphate groups

☒ **C** glycerol attached to three fatty acids and one phosphate group

☒ **D** glycerol attached to three fatty acids and two phosphate groups

(e) The length of a cell was measured. It was found to be 0.0085 cm long.

The ratio of the length of this cell to the length of a chloroplast was 11.7 : 1.

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Calculate the length of this chloroplast.

Give your answer in µm to 2 significant figures.

(2)

Answer ..... µm

(Total for question = 7 marks)

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Q13.

(a) Explain the role of the products of the light-dependent reactions of photosynthesis in the Calvin cycle.

(3)

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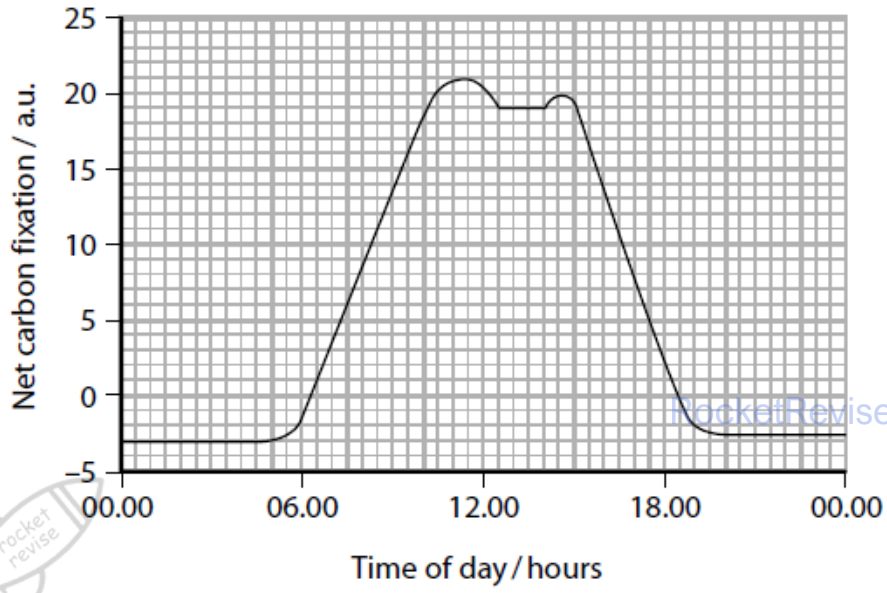
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(b) The graph shows the net carbon fixation in a plant over a period of 24 hours.



(i) Calculate the rate of decrease of net carbon fixation at 12 00 hours.

Include units in your answer.

(3)

Answer .....

(ii) Suggest an explanation for the decrease in net carbon fixation between 12 00 hours and 13 00 hours.

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(3)

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(iii) Suggest why the net carbon fixation was negative before 05 00 hours and after 19 00 hours.

(3)

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(Total for question = 12 marks)

Q14.

Increased levels of atmospheric carbon dioxide can lead to an increased rate of photosynthesis. This is called the CO_2 fertilisation effect.

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This effect varies depending on air and soil temperature, the availability of water, the availability of nutrients and the species of plant.

Plants that can take advantage of higher carbon dioxide levels will have a greater increase in biomass.

(a) Describe how carbon becomes incorporated into plant biomass.

(3)

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(b) (i) Explain why air temperature affects the CO_2 fertilisation effect.

(2)

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(ii) Suggest why soil temperature affects the CO_2 fertilisation effect.

(2)

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(c) Explain why a lack of water availability in the leaves would decrease the CO_2 fertilisation effect.

(3)

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(d) The roots of most land plants are associated with microscopic fungi called mycorrhizal fungi.

These fungi provide the plants with nitrate ions.

The effect of two types of mycorrhizal fungi, ECM and AM, on the change in plant biomass was investigated.

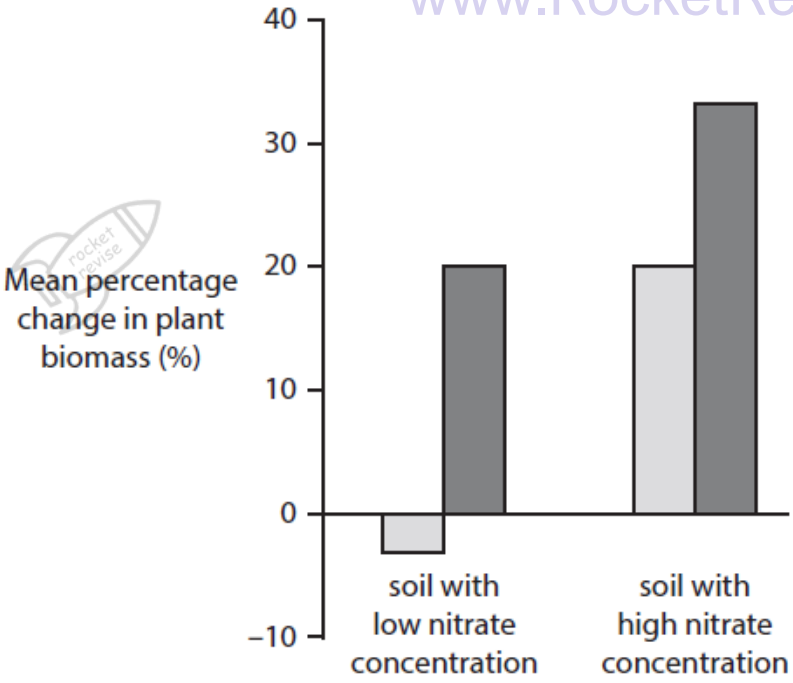
Plants were grown in soil containing either low or high concentrations of nitrate.

The graph shows the results of this investigation.

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Explain the results of this investigation.

(4)

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(Total for question = 14 marks)

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Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Light is essential for photosynthesis.

(a) Explain the role of light in the light-dependent reactions in photosynthesis.

(3)

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(b) The photograph shows thale cress, a small flowering plant found in Europe, Asia and Africa.



(Source: © blickwinkel / Alamy Stock Photo)

The effect of wavelength of light on thale cress was investigated.

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Three groups of plants were treated with a different wavelength of light.

One group of plants was treated with blue light, one group with amber light and one group with red light.

The following were measured:

- rate of photosynthesis
- leaf biomass
- pigment content of the leaves
- changes in transcription of genes in the leaves.

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Another group of these plants had a control light shone on them.

(i) Which would have been the most appropriate light to use as a positive control?

(1)

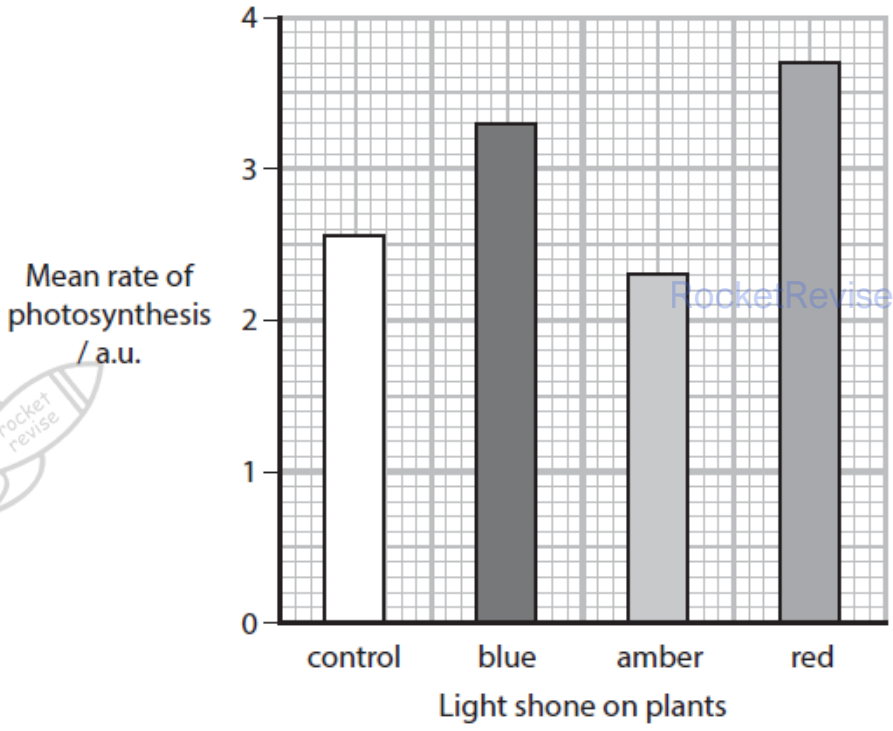
☒ **A** green

☒ **B** mixture of green, blue and red light

☒ **C** ultraviolet

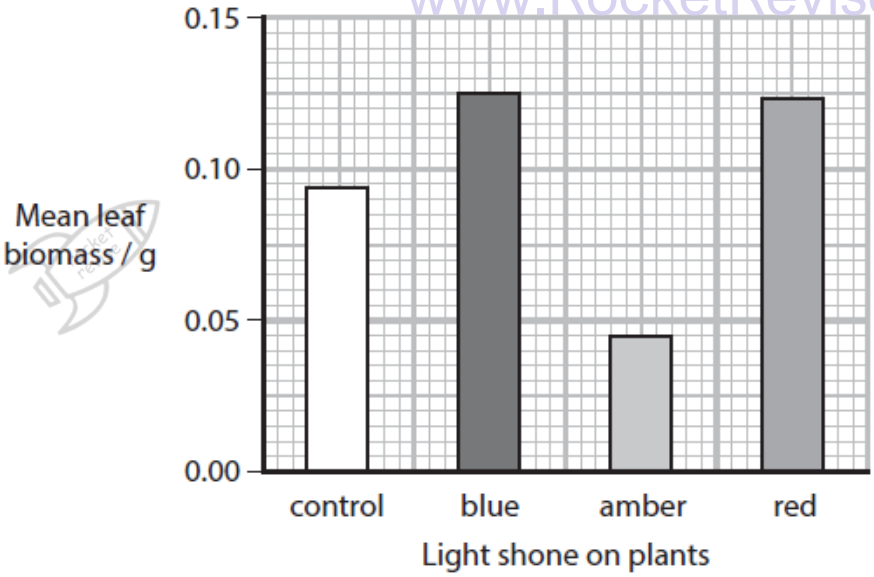
☒ **D** sunlight

(ii) The graphs show the results for the mean rate of photosynthesis and mean leaf biomass.



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Explain the results shown in these graphs.

(4)

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(iii) The table shows the results for the pigment content of the leaves.

Pigment	Mass of pigment / µg per g of leaf biomass			
	Control light	Blue light	Amber light	Red light
Chlorophyll a	382	412	315	441
Chlorophyll b	148	148	125	133
Anthocyanins	78	79	71	72

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Describe **three** conclusions that can be made from these results.

(3)

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(iv) Plants treated with blue light had an increase in photosynthesis caused by an increase in the transcription of some genes in the leaves.

Suggest **two** genes whose transcription could be increased in the presence of blue light.

Give a reason for suggesting each of these genes.

(3)

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(Total for question = 14 marks)

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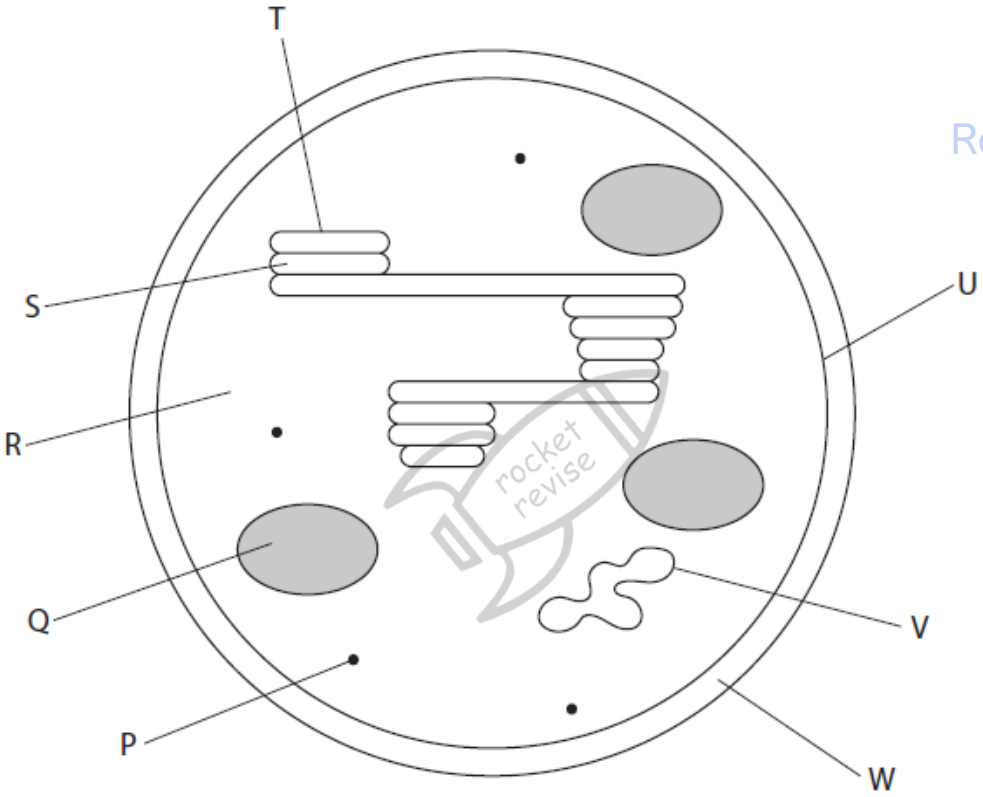
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Q16.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of a chloroplast is related to its role in photosynthesis.

(a) The diagram shows a chloroplast.



(i) Which row of the table shows where the light-dependent reactions and the light-independent reactions take place?

(1)

		Light-dependent reactions	Light-independent reactions
☒	**A**	R	S
☒	**B**	S	T
☒	**C**	T	R
☒	**D**	T	S

(ii) Where do hydrogen ions (protons) accumulate?

(1)

☒ **A** Q

☒ **B** R

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☒ **C** S

☒ **D** W

(iii) The diagram is 16 000 times larger than an actual chloroplast.

Calculate the surface area of the actual chloroplast.

Use the formula:

Surface area = $4\pi r^2$

Give your answer as a whole number in μm^2.

(3)

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Answer ..... μm^2

(iv) Explain the advantages of thylakoids in a chloroplast having a large surface area.

(2)

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(b) In the light-dependent reactions, ATP and reduced NADP are generated.

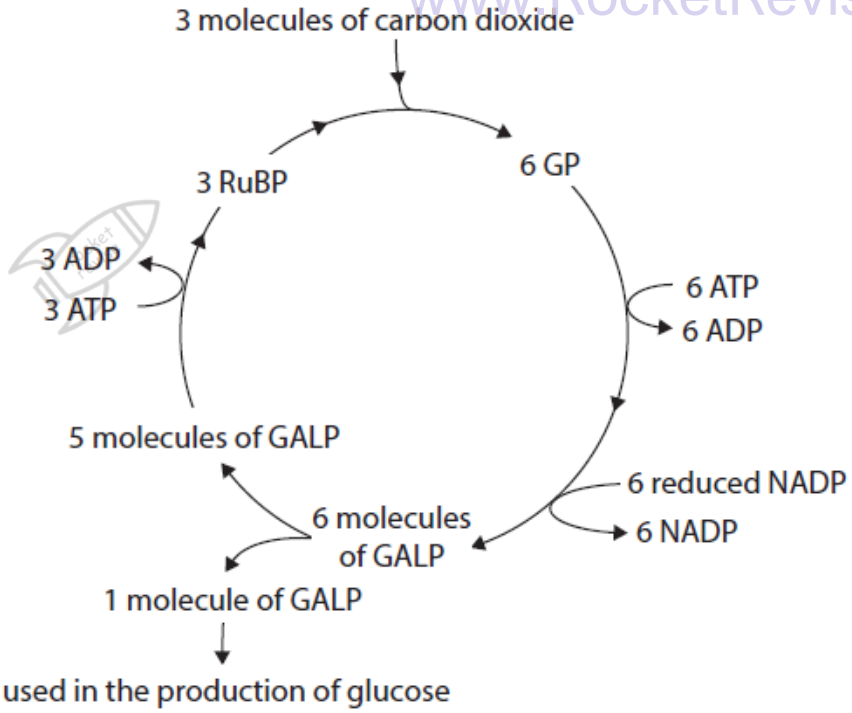
These products are used in the light-independent reactions.

The diagram shows some of the light-independent reactions.

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(i) Which describes how ATP is generated in the light-dependent reactions?

(1)

☒ **A** by hydrolysis of ADP releasing energy

☒ **B** by hydrolysis of ADP requiring energy

☒ **C** by phosphorylation of ADP releasing energy

☒ **D** by phosphorylation of ADP requiring energy

(ii) When one molecule of glucose is made in the light-independent reactions, the ratio of ATP used to reduced NADP used is 1.5 : 1.

Show how this ratio can be calculated, using the information in the diagram.

(2)

(iii) Explain why only a small proportion of the GALP produced is used to make the glucose.

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(2)

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(Total for question = 12 marks)

Q17.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Chloroplasts contain their own DNA.

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(a) (i) Where in the **chloroplast** is this DNA located?

(1)

☒ **A** intergranal membrane

☒ **B** matrix

☒ **C** stroma

☒ **D** thylakoid space

(ii) Which other structures in a plant cell contain DNA?

(1)

☒ **A** amyloplast and mitochondria

☒ **B** mitochondria and nucleus

☒ **C** nucleus and vacuole

☒ **D** vacuole and amyloplast

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(b) The DNA in a chloroplast contains 150 000 **base pairs** in the 105 regions that code for proteins.

(i) Which is the **mean** number of bases in the template (antisense) strand of one region that codes for a protein?

(1)

☒ **A** 239

☒ **B** 476

☒ **C** 714

☒ **D** 1429

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(ii) Suggest **two** reasons why this DNA contains more than 150 000 base pairs.

(2)

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(iii) Describe the roles of the proteins:

- RNA polymerase
- ribosomal protein
- RUBISCO
- ATP synthetase.

(4)

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(Total for question = 9 marks)

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